

# THYROID DIET GUIDELINES

Metabolism-Supporting Nutrition Reference for Hypothyroidism & Hyperthyroidism

## 1. Thyroid Basics

The thyroid gland produces two key hormones: **T3 (triiodothyronine)** and **T4 (thyroxine)**, which regulate basal metabolic rate, temperature, heart rate, digestion, and mood.

| Condition       | Definition                              | Key Symptoms                                                    | Dietary Approach                                                                      |
|-----------------|-----------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Hypothyroidism  | Underactive thyroid; insufficient T3/T4 | Weight gain, fatigue, cold intolerance, constipation, brain fog | Support thyroid hormone synthesis; iodine, selenium, zinc; avoid goitrogens in excess |
| Hyperthyroidism | Overactive thyroid; excess T3/T4        | Weight loss, palpitations, heat intolerance, anxiety, diarrhea  | Adequate calories; calcium; avoid excess iodine; anti-inflammatory focus              |
| Hashimoto's     | Autoimmune hypothyroidism (most common) | Hypothyroid symptoms + immune dysregulation                     | Gluten sensitivity screening; anti-inflammatory diet; selenium priority               |
| Graves' Disease | Autoimmune hyperthyroidism              | Hyperthyroid symptoms + eye changes                             | Avoid iodine-rich supplements; adequate selenium for immune modulation                |

## 2. Key Nutrients for Thyroid Health (CRITICAL)

### 2A. IODINE

Iodine is **essential for T3 and T4 synthesis**. The thyroid gland actively concentrates iodine from the bloodstream. **Recommended daily intake: 150 mcg/day** (adults); 220 mcg (pregnancy); 290 mcg (lactation).

| Iodine Source          | Iodine Content           | Notes                                               |
|------------------------|--------------------------|-----------------------------------------------------|
| Iodized salt           | ~77 mcg per 1/4 tsp      | Most accessible source in India                     |
| Seaweed (nori, wakame) | 16–2984 mcg/g (variable) | Can cause excess in hypothyroidism — use cautiously |
| Cow's milk             | ~56–88 mcg/cup           | Good bioavailable source                            |
| Eggs (whole)           | ~24 mcg per egg          | Regular intake recommended                          |
| Fish (cod, tuna)       | ~63–170 mcg/100g         | Good source; ocean fish richer                      |
| Yogurt (plain)         | ~75 mcg/cup              | Good daily source                                   |
| Prawn/shrimp           | ~35 mcg/100g             | Include in non-veg diet                             |

## 2B. SELENIUM

Selenium is required for the enzymes that **convert T4 (inactive) to T3 (active)** (deiodinases) and protect the thyroid from oxidative damage. **Deficiency worsens Hashimoto's and impairs T3 production.** RDA: 55–70 mcg/day. Evidence: Selenium supplementation (200 mcg/day) significantly reduces thyroid peroxidase antibodies (TPOAb) in Hashimoto's.

| Selenium Source     | Selenium Content   | Notes                                                |
|---------------------|--------------------|------------------------------------------------------|
| Brazil nuts         | ~70–90 mcg per nut | CAUTION: 1–2 nuts/day is sufficient; excess is toxic |
| Tuna (canned)       | ~68 mcg/100g       | Excellent source                                     |
| Sardines            | ~45 mcg/100g       | Good source; omega-3 bonus                           |
| Eggs (whole)        | ~20 mcg per egg    | Include 1–2 daily                                    |
| Sunflower seeds     | ~18 mcg/30g        | Good vegetarian source                               |
| Brown rice (cooked) | ~12 mcg/cup        | Modest source; include daily                         |
| Spinach (cooked)    | ~10 mcg/cup        | Moderate source                                      |

## 2C. ZINC

Zinc is required for TRH (thyrotropin-releasing hormone) synthesis and thyroid receptor function. Zinc deficiency reduces T3 levels and impairs conversion of T4→T3. **RDA: 8–11 mg/day.** Zinc also modulates thyroid autoimmunity in Hashimoto's.

- Pumpkin seeds: ~2.2 mg/30g (excellent vegetarian source)
- Sesame seeds / tahini: ~2.8 mg/30g
- Cashews: ~1.6 mg/30g
- Lentils: ~2.5 mg/cup cooked
- Chicken (dark meat): ~4.0 mg/100g
- Beef: ~4.5 mg/100g
- Fortified cereals: ~3–5 mg/serving

## 3. Protein & Metabolism Support

Hypothyroidism slows metabolism and increases muscle protein breakdown. **Adequate protein intake (1.0–1.5 g/kg/day)** preserves muscle mass and supports thyroid hormone receptor activity. Protein also promotes satiety, which counters hypothyroid-associated weight gain.

## 4. Goitrogens — Safe Consumption Rules

GOITROGENS are naturally occurring compounds in certain foods that can interfere with iodine uptake by the thyroid gland. They are a concern **ONLY** when: (a) iodine intake is already deficient, or (b) consumed in very large quantities (especially raw). Cooking reduces goitrogenic activity by 30–60%. Individuals with adequately treated hypothyroidism and sufficient iodine intake do **NOT** need to eliminate these foods.

| Goitrogenic Food               | Category               | Risk Level        | Safe Consumption Guideline                                                       |
|--------------------------------|------------------------|-------------------|----------------------------------------------------------------------------------|
| Broccoli, cauliflower, cabbage | Cruciferous vegetables | LOW (when cooked) | Eat cooked; 1–2 servings/day is fine                                             |
| Kale, collard greens           | Cruciferous vegetables | LOW (when cooked) | Steam or stir-fry; limit raw in large quantities                                 |
| Soy (tofu, soy milk)           | Legumes                | MODERATE          | Avoid taking soy within 4 hrs of thyroid medication; 1–2 servings/day acceptable |
| Millet (bajra, jowar)          | Grains                 | LOW-MODERATE      | Safe in moderate amounts if iodine intake is adequate; avoid if iodine-deficient |
| Peanuts                        | Legumes                | LOW               | 1–2 tbsp peanut butter/day is safe                                               |
| Sweet potato                   | Root vegetable         | VERY LOW          | No significant concern in cooked form                                            |

## 5. Foods to Prefer

- Iodine-rich: iodized salt, seafood, dairy, eggs
- Selenium-rich: Brazil nuts (1–2/day), tuna, sunflower seeds
- Zinc-rich: pumpkin seeds, lentils, lean meat
- Antioxidant-rich: berries, amla (Vit C), colorful vegetables
- Anti-inflammatory: fatty fish, turmeric, ginger, olive oil
- Fiber-rich: oats, brown rice, vegetables, fruits (for constipation in hypothyroidism)
- Fermented foods: yogurt, kefir, idli/dosa batter — support gut health and immune regulation

## 6. Foods to Limit/Avoid

- **Excessive soy products:** May inhibit thyroid hormone absorption; avoid within 4 hours of levothyroxine.
- **Highly processed foods:** Inflammatory; impair nutrient absorption.
- **Gluten (in Hashimoto's):** 20–30% of Hashimoto's patients show molecular mimicry with gliadin — trial gluten-free for 3 months and assess.
- **Excess raw cruciferous vegetables:** Large daily quantities of raw kale/broccoli can reduce iodine uptake.
- **Excessive iodine supplements (in Hashimoto's/Graves'):** Can trigger thyroid flare; avoid kelp/seaweed supplements unless prescribed.
- **Calcium and iron supplements:** Do NOT take within 4 hours of thyroid medication (impair absorption).
- **High-fat, fried foods:** Reduce thyroid hormone secretion and impair T4-T3 conversion.

## 7. Meal Timing

**Levothyroxine timing rule (CRITICAL for prescribed patients):** Take medication on an empty stomach, 30–60 minutes before breakfast. Avoid coffee, calcium supplements, iron supplements, and high-fiber meals immediately after medication.

- Eat 3 consistent meals per day — metabolic consistency supports thyroid hormone regulation.

- Avoid prolonged calorie restriction (<1200 kcal/day) — can further suppress thyroid function.
- Spacing meals 4–5 hours apart allows adequate T3 action between meals.

## 8. Sample Thyroid-Supportive Diet Plan (1800 kcal)

| Meal             | Time     | Items                                                                 | Thyroid Benefit          |
|------------------|----------|-----------------------------------------------------------------------|--------------------------|
| Before Breakfast | 6:30 AM  | Thyroid medication (if prescribed) + water                            | Optimal drug absorption  |
| Breakfast        | 7:30 AM  | Oats (50g) with egg white omelette (2 eggs) + 1 Brazil nut            | Se, Zn, protein, fiber   |
| Mid-Morning      | 11:00 AM | 1 cup yogurt (plain) + 1 tbsp pumpkin seeds                           | Iodine, zinc, probiotics |
| Lunch            | 1:30 PM  | Brown rice (1 cup) + fish curry / dal + sauteed greens + iodized salt | Iodine, Se, omega-3      |
| Evening          | 5:00 PM  | Sunflower seed chikki or roasted seeds + green tea                    | Selenium, antioxidants   |
| Dinner           | 7:30 PM  | 2 whole wheat rotis + rajma + cooked broccoli + curd                  | Protein, Zn, calcium     |

References: American Thyroid Association Guidelines 2023 • Thyroid Journal Nutrition Reviews • Endocrine Practice 2022 • ICMR RDA 2020